

# MongoDB Sharding

Hands - On - 3

## Objective

The objective of this exercise is to illustrate the concept of sharding, a database partitioning technique for storing large data collections across multiple database servers. For this purpose, you will work with MongoDB, a document-oriented database management system supporting different sharding strategies.

## Material

- Hands on material: <https://github.com/javieraespinosa/dxlab-sharding>
- Option 1 (preferred): play-with-docker.com
- Option 2: Docker Toolbox (<https://www.docker.com/products/docker-toolbox>) or Docker CE (<https://store.docker.com/search?offering=community&q=&type=edition>) depending on your OS

## ToDo

- Access the hands-on material.
- To start follow the instructions in the live session.
- Prepare a report answering the questions proposed in the hands-on material while you perform the simple guided tasks presented in the data lab session.

## Debate on data exploration Material

Consider that you must develop a “vademecum” for helping decision making processes related to data sharding using MongoDB. Use the following questions to interact with the members of your group to develop a set of hints that would contain your vademecum.

- What is a sharding key? Is the choice of a sharding key directly dependent of the sharding strategy? Explain and give examples.
- Explain how does apparently mongodb choses the number of intervals used to shard a collection in the Interval oriented strategy? In which situations would such a strategy be well adapted for sharding a collection?
- In which situations would the hash-based strategy be interesting for a collection to be sharded?
- Which of the strategies interval or sharding would lead to a more balanced distribution of data across shards, interval, or hash?
- What are the advantages and disadvantages of allowing access to shards directly through their server and not only through the query router?
- Give an example of situation where tag based sharding would be an interesting option?
- What happens when a new shard is added to a cluster containing already other shards with data?
- How would you test whether a sharded collection was an interesting solution in comparison to a centralized one?

### ToHand In

- Prepare a document / slides summarizing your examples and discussion results.
- Produce a PDF file of your report, name the file `lis4102-<team number>-ho3.pdf`.